



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to R.N.K FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

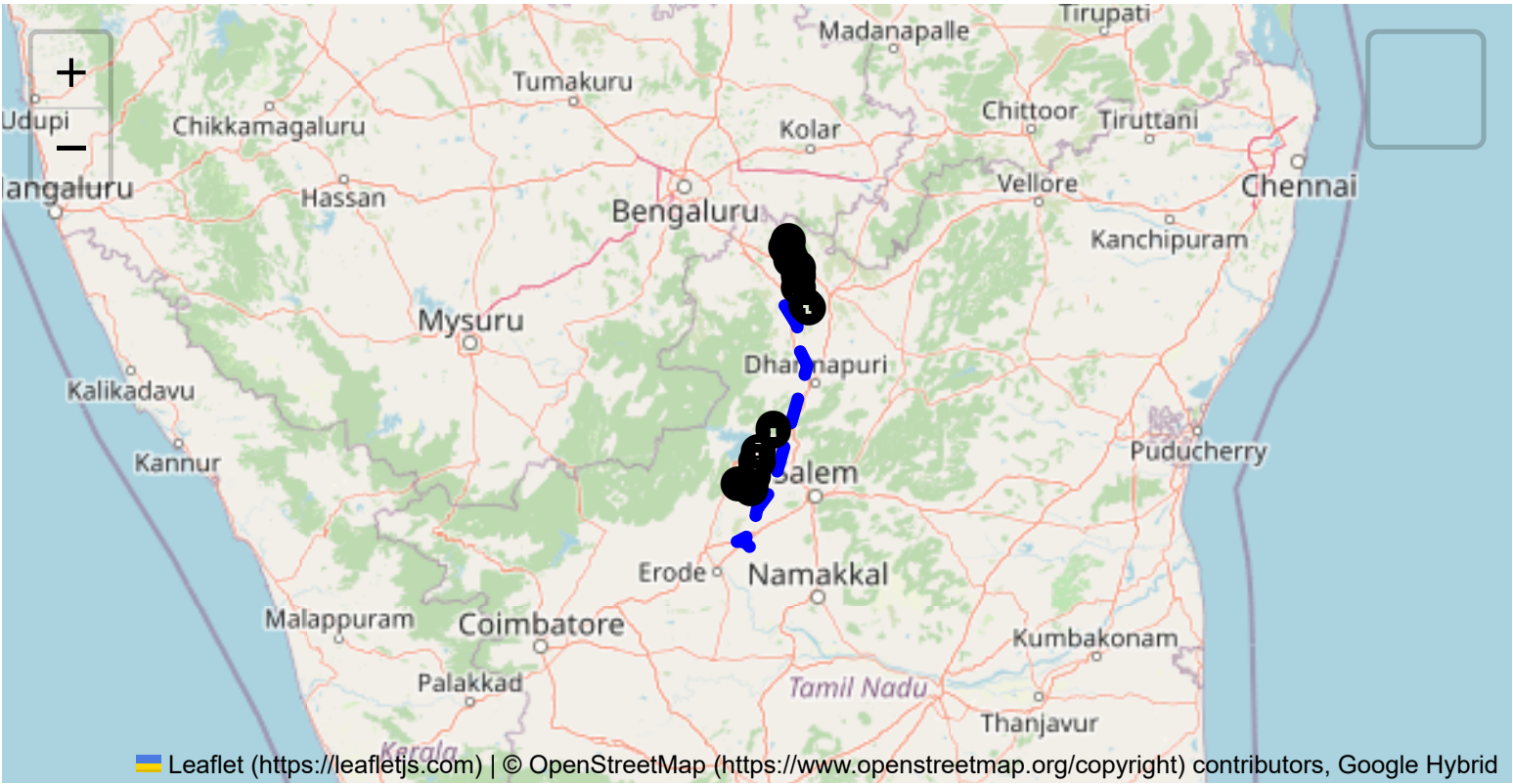
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

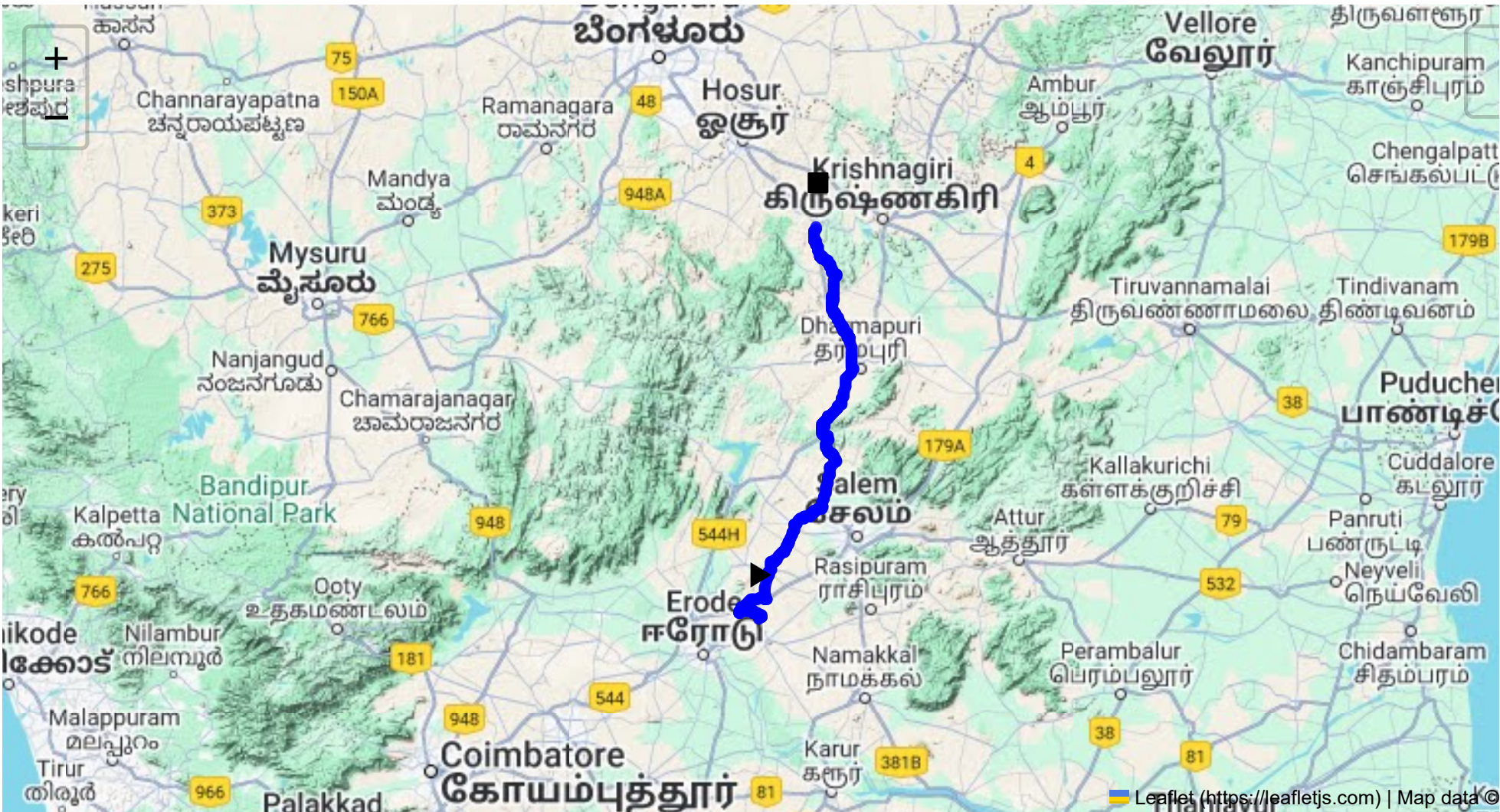
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 155.50 km
Estimated Duration: 3.4 hours
Adjusted Duration (Heavy Vehicle): 4.3 hours
Start: (11.4381, 77.8734)
End: (12.503987, 78.031157)



Welcome to the Journey Risk Management Study

To provide a comprehensive safety analysis of the given route from Sangagiri to Rayakottai via multiple waypoints, we must consider various aspects that can impact the transit of heavy vehicles, especially those carrying hazardous materials. Below is a detailed report based on your requirements:

1. Overview of the Route Map:

- The route stretches approximately 155.50 kilometers, starting from Sangagiri and concluding in Rayakottai, passing through areas like Sankari, Salem, Dharmapuri, and Palacode.
- It involves a combination of state highways (SH 86 and SH 17) and main roads, traversing urban, semi-urban, and rural regions.

2. Typical Weather Conditions and Potential Weather-Related Hazards:

- The region experiences a tropical climate with significant rainfall during the southwest monsoon (June to September) and the northeast monsoon (October to December).
- Potential weather hazards include heavy rainfall leading to waterlogging, reduced visibility, and slippery roads, especially during monsoon seasons.

3. Analysis of Traffic Patterns:

- Urban areas like Salem and proximal regions can experience peak hour congestion, usually in the morning (8 AM - 10 AM) and late afternoon (5 PM - 7 PM).
- Smaller towns along the route may see less congestion but can have localized busy periods, often around market areas or roadway junctions.

4. Assessment of Road Quality and Infrastructure:

- State highways are generally well-maintained, but local roads might present issues like potholes or uneven surfaces.
- Bridges and underpasses should be reviewed for weight limitations that might be pertinent to heavy or oversized vehicles.

5. Suggestions for Alternative Routes for Emergencies:

- An alternate route could involve using NH 44 (part of India's Golden Quadrilateral) for faster and potentially safer transit, though it may require deviations.
- The National Highways Authority of India (NHAI) app or local Google Maps traffic layer can offer real-time rerouting suggestions during transit.

6. Summary of Local Regulations Affecting Hazardous Material Transport:

- The transport of hazardous materials in Tamil Nadu is governed by the Motor Vehicles Act and several state-specific guidelines. Compliance with mandatory signage, documentation, and permits is crucial.
- Night travel for hazardous materials might be restricted, requiring permits.

7. Overview of Historical Incidents:

- No significant publicized incidents specific to hazardous material transport were reported historically along this route, but caution and preparedness for possible emergencies are advised.

8. Environmental Considerations and Sensitive Areas:

- The route passes near agricultural land and small water bodies, hence risk assessment should include potential spill impacts on local ecosystems.
- Avoidance of sensitive zones like schools and hospitals during peak hours is recommended.

9. Analysis of Communication Coverage:

- While major towns should have strong cellular and data coverage, rural segments might have sporadic dead zones.

- It’s advisable for drivers to have alternate communication methods, like satellite phones, for emergencies.

10. Estimated Emergency Response Times:

- Urban areas: 15-30 minutes, thanks to proximity to emergency services.
- Rural areas: Could be 30-60 minutes depending on road conditions and the remoteness.
- Local coordination with police and fire stations en route is advised for planning and quicker response.

11. Overall Summary of Risk Assessment:

- While the primary route is feasible, attention must be paid to monsoon-related hazards, potential road quality issues, local traffic regulations, and emergency preparedness.
- Training with a focus on defensive driving, familiarity with the route, and emergency drills can mitigate potential risks significantly.
- Continuous monitoring of traffic and weather updates will be essential to maintain a high safety standard for transporting hazardous materials.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43968, 77.87345	15 KM/Hr
1	Turn	High	11.44029, 77.87544	15 KM/Hr
2	Turn	High	11.44900, 77.87406	15 KM/Hr
3	Turn	High	11.45352, 77.85698	15 KM/Hr
4	Turn	Medium	11.45764, 77.85219	30 KM/Hr
5	Turn	High	11.46303, 77.84945	15 KM/Hr
6	Turn	High	11.45571, 77.81702	15 KM/Hr
7	Turn	Medium	11.45654, 77.81706	30 KM/Hr
8	Turn	Medium	11.46916, 77.82431	30 KM/Hr
9	Blind Spot	Blind Spot	11.49225, 77.89560	10 KM/Hr
10	Turn	High	11.49453, 77.88488	15 KM/Hr
11	Turn	Medium	11.56348, 77.90077	30 KM/Hr
12	Turn	Medium	11.57547, 77.90567	30 KM/Hr
13	Turn	Medium	11.59189, 77.90645	30 KM/Hr
14	Turn	Medium	11.59323, 77.90796	30 KM/Hr
15	Turn	Medium	11.63857, 77.94831	30 KM/Hr
16	Turn	High	11.64207, 77.95199	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
17	Turn	Medium	11.64708, 77.95119	30 KM/Hr
18	Turn	High	11.69596, 77.97122	15 KM/Hr
19	Turn	Medium	11.71721, 77.99982	30 KM/Hr
20	Turn	Medium	11.71923, 78.01140	30 KM/Hr
21	Turn	Medium	11.73772, 78.03606	30 KM/Hr
22	Turn	High	11.73665, 78.03907	15 KM/Hr
23	Turn	Medium	11.74226, 78.04688	30 KM/Hr
24	Turn	Medium	11.77961, 78.05775	30 KM/Hr
25	Turn	Medium	11.86873, 78.08374	30 KM/Hr
26	Turn	Medium	11.87966, 78.08187	30 KM/Hr
27	Turn	Medium	11.91211, 78.05852	30 KM/Hr
28	Turn	Medium	11.92609, 78.06513	30 KM/Hr
29	Turn	Medium	11.93334, 78.06409	30 KM/Hr
30	Turn	Medium	11.93719, 78.05372	30 KM/Hr
31	Turn	Medium	11.95457, 78.04685	30 KM/Hr
32	Turn	Medium	11.95688, 78.04803	30 KM/Hr
33	Turn	High	11.95828, 78.04983	15 KM/Hr
34	Turn	Medium	11.95997, 78.04946	30 KM/Hr
35	Turn	Medium	11.96335, 78.05141	30 KM/Hr
36	Turn	High	11.96408, 78.05131	15 KM/Hr
37	Turn	Medium	11.96511, 78.04966	30 KM/Hr
38	Turn	High	11.96608, 78.04919	15 KM/Hr
39	Turn	Medium	11.96819, 78.05088	30 KM/Hr
40	Turn	Medium	12.12010, 78.13282	30 KM/Hr
41	Turn	High	12.20914, 78.11681	15 KM/Hr
42	Blind Spot	Blind Spot	12.20970, 78.11680	10 KM/Hr
43	Blind Spot	Blind Spot	12.20378, 78.12304	10 KM/Hr
44	Turn	Medium	12.22451, 78.10903	30 KM/Hr
45	Turn	Medium	12.23481, 78.10507	30 KM/Hr
46	Turn	Medium	12.23511, 78.10444	30 KM/Hr
47	Turn	Medium	12.26532, 78.08427	30 KM/Hr
48	Turn	High	12.29602, 78.07407	15 KM/Hr
49	Turn	Medium	12.29641, 78.07527	30 KM/Hr
50	Turn	High	12.31180, 78.07854	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
51	Blind Spot	Blind Spot	12.31360, 78.07464	10 KM/Hr
52	Turn	High	12.37241, 78.08518	15 KM/Hr
53	Turn	Medium	12.37573, 78.08362	30 KM/Hr
54	Turn	High	12.37875, 78.08503	15 KM/Hr
55	Turn	Medium	12.38703, 78.07477	30 KM/Hr
56	Turn	Medium	12.39347, 78.07169	30 KM/Hr
57	Turn	Medium	12.40605, 78.06717	30 KM/Hr
58	Turn	Medium	12.41316, 78.06064	30 KM/Hr
59	Turn	Medium	12.41690, 78.05592	30 KM/Hr
60	Turn	Medium	12.41829, 78.05120	30 KM/Hr
61	Blind Spot	Blind Spot	12.42547, 78.04645	10 KM/Hr
62	Turn	High	12.42472, 78.04656	15 KM/Hr
63	Turn	Medium	12.42292, 78.04507	30 KM/Hr
64	Turn	Medium	12.42274, 78.04450	30 KM/Hr
65	Blind Spot	Blind Spot	12.42350, 78.04307	10 KM/Hr
66	Turn	Medium	12.42376, 78.04341	30 KM/Hr
67	Turn	Medium	12.42677, 78.04418	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Sri Shanthi Hospital	11.569272, 77.901937	30 km/h	Medium
2	clinic	Sun Eye Care	11.6936242, 77.9730948	30 km/h	Medium
3	hospital	Dr. Pandiyarajan Hospital	11.692704, 77.969344	30 km/h	Medium
4	hospital	Dr. Sengodan Hospital	11.6959527, 77.9695325	30 km/h	Medium
5	hospital	Omalur Government Hospital	11.74161, 78.044296	30 km/h	Medium
6	clinic	R.R. Nursing Home	11.7394129, 78.0439861	30 km/h	Medium
7	hospital	Sri Hari Hospital	11.7396726, 78.0440701	30 km/h	Medium
8	hospital	Sugumar Hospital	11.7396031, 78.0426955	30 km/h	Medium
9	hospital	Sri Sugam Hospital	11.7429821, 78.0461039	30 km/h	Medium
10	hospital	Saravana Hospital	11.7409303, 78.0458308	30 km/h	Medium
11	hospital	Arun Hospital	11.7429892, 78.0466881	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
12	clinic	Saratha Clinic	11.7422818, 78.0458175	30 km/h	Medium
13	clinic	Sri Meenakshi Nursing Home, Salem	11.7449162, 78.0474637	30 km/h	Medium
14	clinic	Divyam Child Care Clinic	11.7447135, 78.0473858	30 km/h	Medium
15	clinic	Gomathi Clinic	11.7445293, 78.0470401	30 km/h	Medium
17	hospital	Vinayaka Child Hospital	11.745812, 78.0480225	30 km/h	Medium
18	clinic	Momota Clinic	11.744982, 78.047397	30 km/h	Medium
19	hospital	Amirthalingam Hospital	11.7456708, 78.0470548	30 km/h	Medium
20	clinic	Shri Ramana Poly Clinic	11.7477122, 78.0479293	30 km/h	Medium
21	hospital	Dr. G.R.Thilakam Hospital	11.746097, 78.04769	30 km/h	Medium
22	hospital	Dr. Nagaraj Hospital	11.7464035, 78.0472401	30 km/h	Medium
23	hospital	Thulasi Hospital	11.7461516, 78.0469984	30 km/h	Medium
24	hospital	Children Hospital	11.7458377, 78.0480205	30 km/h	Medium
25	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
26	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
28	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
29	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
30	hospital	Thoppur,Government Hospital	11.9398824, 78.0551658	30 km/h	Medium
32	hospital	Government Hospital Palacode	12.2964824, 78.0761735	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
16	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
27	college	Dhirajlal Gandhi College of Technology	11.7968014, 78.0562095	30 km/h	Medium
31	college	Laxminarayana Polytechnic College	12.0421277, 78.103203	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45352, 77.85698



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45764, 77.85219



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46303, 77.84945



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45571, 77.81702



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45654, 77.81706



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46916, 77.82431



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.49225, 77.89560



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49453, 77.88488



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.56348, 77.90077



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.57547, 77.90567



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.59189, 77.90645



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.59323, 77.90796



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.63857, 77.94831



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.64207, 77.95199



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64708, 77.95119



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.69596, 77.97122



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71721, 77.99982



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71923, 78.01140



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.73772, 78.03606



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.73665, 78.03907



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.74226, 78.04688



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.77961, 78.05775



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.86873, 78.08374



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.87966, 78.08187



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.91211, 78.05852



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.92609, 78.06513



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93334, 78.06409



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93719, 78.05372



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95457, 78.04685



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95688, 78.04803



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.95828, 78.04983



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95997, 78.04946



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96335, 78.05141



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96408, 78.05131



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96511, 78.04966



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96608, 78.04919



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96819, 78.05088



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.12010, 78.13282



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.20914, 78.11681



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 12.20970, 78.11680



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.22451, 78.10903



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.23511, 78.10444



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.26532, 78.08427



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.29602, 78.07407



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.29641, 78.07527



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.31180, 78.07854



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 12.31360, 78.07464



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.37241, 78.08518



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.37573, 78.08362



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.37875, 78.08503



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.38703, 78.07477



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.39347, 78.07169



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.40605, 78.06717



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.41316, 78.06064



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.41690, 78.05592



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.41829, 78.05120



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 12.42547, 78.04645



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.42472, 78.04656



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.42292, 78.04507



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.42274, 78.04450

